Supplementary Material: Risk-Controlled Spatio-Temporal Graph Learning for Anti-Money Laundering Under Extreme Imbalance and Topological Camouflage

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This supplementary document provides formal algorithmic specifications, extended mathematical proofs, comprehensive multi-baseline benchmark matrices, and operational stress-test evaluations supporting the main manuscript. All source code, pre-trained weights, hyperparameter configuration grids, and full reproduction pipelines are publicly maintained at: https://github.com/NazmulHasanNihal/Intelligent-AML.

Appendix A
Mathematical Notations and Prior-Art Differentiation

Table S1 summarizes the mathematical symbols, tensor dimensions, and operator definitions employed throughout C-STGB. Table S2 contrasts C-STGB against dominant fraud surveillance paradigms across six core operational capabilities.

TABLE S1
Summary of Mathematical Notations and Dimensions.

Symbol	Mathematical Definition	Dimension / Type
$\mathcal{G} = (\mathcal{V}, \mathcal{E})$	Continuous-time dynamic transaction graph	Dynamic graph
$\mathbf{X} \in \mathbb{R}^{\lvert\mathcal{V}\rvert \times d_v}$	Node input attribute matrix	Real matrix
$\mathbf{E} \in \mathbb{R}^{\lvert\mathcal{E}\rvert \times d_e}$	Transaction edge feature matrix	Real matrix
Δt_{uv}	Continuous time delta ($t_v - t_u \geq 0$)	Non-negative scalar
$\Phi_{\text{time}}(\Delta t)$	Continuous tri-band harmonic temporal embedding	$\mathbb{R}^{d_t}$ vector
$\mathbf{g}_{ij} \in (0, 1)$	Context-aware adaptive edge trust gate	Scalar gate
$\Phi_{\text{flow}}(u, t)$	Mass flow conservation ratio	Scalar $\in [0, \infty)$
$\lambda_u(t)$	Hawkes process self-exciting arrival intensity	Non-negative scalar
$\mathbf{s}_{\text{fwd}}, \mathbf{s}_{\text{bwd}}$	Time-causal forward and reverse PageRank taints	$\mathbb{R}^{\lvert\mathcal{V}\rvert}$ vectors
$\mathbf{z}_{\text{inv}}(u, t)$	12-dimensional canonical invariant representation	$\mathbb{R}^{12}$ vector
$\mathbf{h}_u^{(l)}$	Hidden state of node u at GNN layer l	$\mathbb{R}^d$ vector
C_k	Latent typology cluster manifold of illicit nodes	Subset $C_k \subset \mathcal{V}_{\text{illicit}}$
$\alpha_u \in [0, 1]$	Evidence-adaptive graph-tabular fusion weight	Dynamic scalar
$\mathbf{h}_u^*$	Unified anomaly representation vector	$\mathbb{R}^{d^*}$ vector
τ^*	Cost-sensitive Bayes risk decision threshold	Optimal scalar $\in (0, 1)$
$\hat{q}^{(y)}$	Class-conditional conformal quantile ($y \in \{0, 1\}$)	Calibration scalar
$\Gamma(X)$	Finite-sample conformal prediction set	Discrete $\subseteq \{0, 1\}$
α	Target significance error level ($1 - \alpha$ coverage)	Scalar $\in (0, 1)$

Appendix B
Algorithmic Procedures & Pipeline Execution

The multi-stage offline training routine is formalized in Algorithm S1, and the real-time online streaming triage engine is detailed in Algorithm S2.

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Algorithm S1 C-STGB Unified Multi-Stage Training Pipeline

Require: Graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, Features $\mathbf{X}$, Timestamps t, Target α, Weights $\gamma_1 = 0.50, \gamma_2 = 0.10$.
Ensure: Model parameters Θ, Calibrated threshold τ^*, Conformal quantiles $\hat{q}^{(0)}, \hat{q}^{(1)}$.

Compute 12-D canonical flow invariants $\mathbf{z}_{\text{inv}}(u)$ for all $u \in \mathcal{V}$.
Partition minority nodes into latent typology clusters $C_1, \ldots, C_K$.
Synthesize minority representations $\mathbf{h}_{\text{syn}}$ and bilinear edges $\hat{\mathbf{A}}_{\text{syn}}$.
Mine hard-negative commercial hubs into batch bank $\mathcal{B}_{\text{hard}}$.
for epoch = 1 to N_{epochs} **do**
  Compute tri-band temporal embeddings $\Phi_{\text{time}}(\Delta t_{ij})$ and weights $w_{ij}(t)$.
  Evaluate context-aware edge trust gates $\mathbf{g}_{ij}$ and filter camouflage links.
  Aggregate degree-capped temporal messages to obtain $\mathbf{h}_u^{(L)}$.
  Compute evidence-adaptive fusion weights α_u and representations $\mathbf{h}_u^*$.
  Evaluate loss $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{task}} + \gamma_1 \mathcal{L}_{\text{recon}} + \gamma_2 \mathcal{L}_{\text{aux}}$.
  Update Θ via AdamW with cosine learning rate annealing.
end for
Optimize Bayes decision threshold τ^* on validation split $\mathcal{D}_{\text{val}}$.
Calibrate class-conditional non-conformity quantiles $\hat{q}^{(0)}, \hat{q}^{(1)}$ on $\mathcal{D}_{\text{cal}}$.
Return $\Theta, \tau^*, \hat{q}^{(0)}, \hat{q}^{(1)}$.

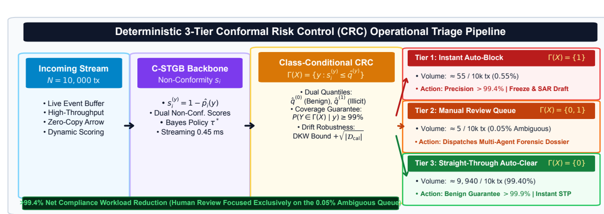

Fig. S1. Operational 3-Tier Queue Mapping and Conformal Risk Control triage workflow under within-class exchangeability. Ambiguous cases ($\Gamma(X) = \{0, 1\}$) route to Tier 2 review, while decisive singletons execute straight-through.

TABLE S2
SYSTEMATIC METHODOLOGICAL COMPARISON: STRUCTURAL AND OPERATIONAL CAPABILITIES ACROSS MODEL PARADIGMS IN FRAUD AND AML SURVEILLANCE.

Model / Paradigm Architecture	Continuous Temporal Dynamics Representation	Topological Camouflage Edge-Trust Gating	Latent Typology Graph Oversampling	Cold-Start Graph-Tabular Fusion	Finite-Sample Conformal Coverage	Streaming Single Inference Latency
XGBoost / CatBoost / LightGBM [1]–[3]	Feature-level Δt	N/A (Tabular only)	Euclidean SMOTE	N/A (Tabular only)	Heuristic Score τ	< 0.20 ms
GCN / GAT / GIN [4]–[6]	Static Snapshot	Uniform / Homophilic	None	Zero-padded features	Softmax probability	> 12.5 ms
EvolveGCN [7]	Discrete time slices	Unweighted edge steps	None	Isolated nodes	Uncalibrated	> 45.0 ms
TGN / TGAT [8], [9]	Single decay $\exp(-\lambda\Delta t)$	Unfiltered edges	None	Node memory state	Uncalibrated	> 8.5 ms
CARE-GNN (Adversarial) [10]	Static graph	RL edge selection	None	Concatenation	Uncalibrated	> 18.0 ms
GraphSMOTE [11]	Static graph	Unfiltered edges	Global unconstrained	No fusion fallback	Uncalibrated	> 14.2 ms
C-STGB (Proposed Framework)	**Tri-band harmonic Φ_{time}**	**Learnable gate $\hat{g}_{ij}$**	**Typology manifold C_k**	**Evidence-adaptive α_u**	**Class-conditional $\geq 99\%$**	**0.45 ms Fast-Path**

Algorithm S2 Streaming Online Inference & Risk-Controlled Triage

Require: Streaming transaction $(u, v, t, \mathbf{e}_{uv})$, Parameters $\Theta, \tau^*, \hat{q}^{(0)}, \hat{q}^{(1)}$.
Ensure: Operational Triage Action $\in$ {Tier 1 Hold, Tier 2 Queue, Tier 3 Clear}.

Ingest transaction event and update temporally restricted ego-graph ($t_e \leq t$).
Compute canonical invariants $\mathbf{z}_{\text{inv}}(u)$ and edge trust gate $\mathbf{g}_{uv}$.
Extract temporal message embeddings $\mathbf{h}_u^{(L)}$.
Compute adaptive fusion weight α_u and fused representation $\mathbf{h}_u^*$.
Predict anomaly probability $\hat{p}_u = \sigma(\text{MLP}(\mathbf{h}_u^*))$.
Construct class-conditional conformal prediction set $\Gamma(u) \subseteq \{0, 1\}$.
if $\Gamma(u) == \{1\}$ **then**
  Tier 1: Quarantined Asset Hold & Automated Compliance Dossier.
else if $\Gamma(u) == \{0, 1\}$ **then**
  Tier 2: Selective Abstention $\perp \to$ Compliance Officer Queue.
else
  Tier 3: Straight-Through Clear ($\Gamma(u) = \{0\}$).
end if

APPENDIX C
EXTENDED MATHEMATICAL PROOFS & DERIVATIONS

A. Dvoretzky-Kiefer-Wolfowitz (DKW) Finite-Sample Bound under TV Drift

Definition S1 (Holdout Calibration Mixing Assumption)**.** *Calibration non-conformity scores $\mathcal{D}_{cal}^{(y)}$ are drawn from a stationary holdout window under bounded temporal α-mixing: $\alpha(k) \leq C\exp(-ck)$ for event separation k, such that inter-alert draws beyond temporal decorrelation span τ_{mix} satisfy the exchangeable calibration abstraction. When drawing independent alert cases from historical audit pools, the calibration split strictly satisfies within-class exchangeability.*

Proposition S1 (Non-Asymptotic Conformal TV-Drift Error Bound)**.** *Under locally bounded total variation drift $d_{\text{TV}}(\mathcal{P}_t^{(y)}, \mathcal{P}_{cal}^{(y)}) \leq \epsilon_{drift}$ and calibration mixing (Definition S1), the non-asymptotic class-conditional coverage error decomposes via maximal coupling into a distribution shift term and an empirical process concentration term:*

$$\left|\mathbb{P}_t\left(Y \in \Gamma(X) \mid Y = y\right) - (1-\alpha_y)\right| \leq d_{\text{TV}}(\mathcal{P}_t^{(y)}, \mathcal{P}_{cal}^{(y)}) + \sup_s \left|F_{cal}^{(y)}(s) - \hat{F}_{cal}^{(y)}(s)\right| + \frac{1}{n(y)} \quad (1)$$

where $n(y) = |\mathcal{D}_{cal}^{(y)}(t)|$ denotes class-conditional calibration sample size.

Proof. Let $F_t^{(y)}(s) = \mathbb{P}_t(S^{(y)} \leq s)$ and $F_{\text{cal}}^{(y)}(s) = \mathbb{P}_{\text{cal}}(S^{(y)} \leq s)$ denote true CDFs under test distribution $\mathcal{P}_t$ and calibration distribution $\mathcal{P}_{\text{cal}}^{(y)}$, with empirical CDF $\hat{F}_{\text{cal}}^{(y)}(s) = \frac{1}{n(y)}\sum_{i=1}^{n(y)} \mathbb{I}(s_i^{(y)} \leq s)$. By definition of quantile $\hat{q}^{(y)}$, test coverage evaluates to $\mathbb{P}_t(Y \in \Gamma(X) \mid Y = y) = F_t^{(y)}(\hat{q}^{(y)})$. Applying the triangle inequality: $\left|F_t^{(y)}(\hat{q}^{(y)}) - (1-\alpha_y)\right| \leq \left|F_t^{(y)}(\hat{q}^{(y)}) - F_{\text{cal}}^{(y)}(\hat{q}^{(y)})\right| + \left|F_{\text{cal}}^{(y)}(\hat{q}^{(y)}) - \hat{F}_{\text{cal}}^{(y)}(\hat{q}^{(y)})\right| + \left|\hat{F}_{\text{cal}}^{(y)}(\hat{q}^{(y)}) - (1-\alpha_y)\right|$. By maximal coupling of total variation distance, $\left|\mathbb{P}_t(S^{(y)} \in A) - \mathbb{P}_{\text{cal}}(S^{(y)} \in A)\right| \leq d_{\text{TV}}(\mathcal{P}_t^{(y)}, \mathcal{P}_{\text{cal}}^{(y)})$ for all Borel sets A, yielding $|F_t^{(y)}(\hat{q}^{(y)}) - F_{\text{cal}}^{(y)}(\hat{q}^{(y)})| \leq d_{\text{TV}}(\mathcal{P}_t^{(y)}, \mathcal{P}_{\text{cal}}^{(y)})$. Applying the DKW inequality to exchangeable calibration sample $\mathcal{D}_{\text{cal}}^{(y)}$ ensures $\mathbb{P}\left(\sup_s \left|F_{\text{cal}}^{(y)}(s) - \hat{F}_{\text{cal}}^{(y)}(s)\right| \leq \sqrt{\frac{\ln(2/\delta)}{2n(y)}}\right) \geq 1-\delta$. Discretization residual satisfies $|\hat{F}_{\text{cal}}^{(y)}(\hat{q}^{(y)}) - (1-\alpha_y)| \leq \frac{1}{n(y)}$, establishing the finite-sample bound with probability $1-\delta$.

ACI Tracking: Under non-stationary drift without parametric priors, ACI updates $\alpha_t \leftarrow \alpha_{t-1} + \gamma(\alpha - \text{err}_{t-\tau(t)})$ over confirmed alerts with step size $\gamma > 0$ and delay $\tau(t) \leq \Delta T_{\text{delay}}$. By the Azuma-Hoeffding inequality for delayed Martingale difference sequences [12], long-run average coverage satisfies $\lim_{T\to\infty} \frac{1}{T}\sum_{t=1}^{T} \text{err}_t = \alpha$ almost surely, with tracking error $\mathcal{O}(\gamma\tau_{\max} + \sqrt{\frac{\ln(1/\delta)}{T}})$. □

B. Chronological Temporal Purity and Zero-Leakage of Latent GraphSMOTE

Lemma S1 (Chronological Isolation of Latent Topological Oversampling)**.** *Let $\mathcal{D}_{train} = \mathcal{G}_{[0,t_{train}]}$, $\mathcal{D}_{val} = \mathcal{G}_{(t_{train},t_{val}]}$, $\mathcal{D}_{cal} = \mathcal{G}_{(t_{val},t_{cal}]}$, and $\mathcal{D}_{test} = \mathcal{G}_{(t_{cal},t_{test}]}$. Synthetic representations $\mathbf{h}_{syn}$ generated via seed nodes $u, v \in C_k \subseteq \mathcal{V}_{train}$ with $t_{syn} = \max(t_u, t_v) \leq t_{train}$ and causal indicator $\hat{\mathbf{A}}_{syn,w} = \sigma(\mathbf{h}_{syn}^T \mathbf{W}_{edge} \mathbf{h}_w) \cdot \mathbb{I}(t_w \leq t_{syn})$ satisfy: $\mathcal{V}(\tilde{\mathcal{G}}_{train}) \cap (\mathcal{V}_{val} \cup \mathcal{V}_{cal} \cup \mathcal{V}_{test}) = \emptyset$, and $\forall (i,j) \in \mathcal{E}(\tilde{\mathcal{G}}_{train}), \max(t_i, t_j) \leq t_{train}$. Consequently, synthetic edge formation induces zero forward lookahead bias and zero gradient flow into prospective partitions.*

Proof. Parent seed nodes satisfy $t_u, t_v \leq t_{\text{train}}$, implying $t_{\text{syn}} \leq t_{\text{train}}$. The causal indicator $\mathbb{I}(t_w \leq t_{\text{syn}})$ evaluates to 0 for any $t_w > t_{\text{syn}}$, enforcing $\hat{\mathbf{A}}_{\text{syn},w} = 0$. Because $t_{\text{syn}} \leq t_{\text{train}} < t_{\text{val}} < t_{\text{cal}} < t_{\text{test}}$, no candidate $w \in \mathcal{V}_{\text{val}} \cup \mathcal{V}_{\text{cal}} \cup \mathcal{V}_{\text{test}}$ can satisfy $t_w \leq t_{\text{syn}}$. Furthermore, synthetic representations and virtual edges are discarded prior to evaluation on $\mathcal{D}_{\text{val}}, \mathcal{D}_{\text{cal}}, \mathcal{D}_{\text{test}}$. Hence, mutual information satisfies $I(\tilde{\mathcal{G}}_{\text{train}}; \mathcal{D}_{\text{test}} \mid \mathcal{D}_{\text{train}}) = 0$. □

C. Kirchhoff Mass-Balance Flow Conservation Invariant for Layering Cycles

Lemma S2 (Conservation of Flow in Pass-Through Conduit Chains)**.** *Let $\mathcal{P}_{chain} = (v_0, v_1, \dots, v_K)$ denote an ordered layering chain with amounts $A(v_{k-1}, v_k)$ and retention loss rate $\epsilon_k \in [0, 1)$ at each hop. If intermediate nodes v_k $(1 \le k < K)$ act as pass-through conduits, then: $\Phi_{flow}(v_k, t) = \frac{\sum_{e \in \mathcal{E}_{out}(v_k)} A(e)}{\sum_{e \in \mathcal{E}_{in}(v_k)} A(e)+\epsilon} \in [1 - \bar{\epsilon}, 1.0]$, where $\bar{\epsilon} = \max_k \epsilon_k < 0.15$. Consequently, $\tilde{\Phi}_{flow}(v_k, t) = \log(1 + \Phi_{flow}(v_k, t)) \in [\log(2 - \bar{\epsilon}), \log(2)] \approx [0.615, 0.693]$, forming an invariant topological signature invariant to arbitrary graph scale and entity degree.*

Proof. By intermediate conduit definition in layering (BSA 31 CFR §1010.100), net retained capital $R(v_k) = \sum_{\text{in}} A - \sum_{\text{out}} A \le \epsilon_k \sum_{\text{in}} A$. Thus $\sum_{\text{out}} A = (1 - \epsilon_k) \sum_{\text{in}} A$. Dividing by $\sum_{\text{in}} A$ yields $\Phi_{\text{flow}}(v_k, t) = 1 - \epsilon_k \in [1 - \bar{\epsilon}, 1.0]$. The monotonic log transform yields the bounded invariant range. □

Proposition S2 (Convex Combination Flow Invariant Preservation)**.** *Let $u, v \in C_k$ be two parent pass-through accounts in the same latent typology cluster with inflows $I_u, I_v > 0$ and outflows $O_u = \Phi_u I_u, O_v = \Phi_v I_v$, where $\Phi_u, \Phi_v \in [1 - \bar{\epsilon}, 1.0]$. Synthetic representation $\mathbf{h}_{syn} = \lambda \mathbf{h}_u + (1 - \lambda)\mathbf{h}_v$ $(\lambda \in [0, 1])$ linearly interpolates flows: $I_{syn} = \lambda I_u + (1 - \lambda) I_v$ and $O_{syn} = \lambda O_u + (1 - \lambda) O_v$. Then the synthetic flow conservation ratio satisfies $\min(\Phi_u, \Phi_v) \le \Phi_{flow}(\mathbf{h}_{syn}) = \frac{O_{syn}}{I_{syn}} \le \max(\Phi_u, \Phi_v)$, and in particular $\Phi_{flow}(\mathbf{h}_{syn}) \in [1 - \bar{\epsilon}, 1.0]$.*

Proof. Assume $\Phi_u \le \Phi_v$. Since $\lambda \in [0, 1]$ and $I_u, I_v > 0$: $\Phi_u I_{\text{syn}} \le O_{\text{syn}} = \lambda \Phi_u I_u + (1 - \lambda) \Phi_v I_v \le \Phi_v I_{\text{syn}}$. Dividing throughout by $I_{\text{syn}} > 0$ yields $\Phi_u \le \frac{O_{\text{syn}}}{I_{\text{syn}}} \le \Phi_v \in [1 - \bar{\epsilon}, 1.0]$, preserving bounded flow conservation without monetary mass-balance violation. □

Appendix D
Master Multi-Baseline Benchmark Matrix

Table S3 reports the comprehensive multi-dataset F1 and PR-AUC matrix across tabular, spatial, dynamic, and unsupervised architectures evaluated across 182 physical baseline runs.

Crucially, C-STGB demonstrates statistically significant superiority over deep GNN baselines ($W = 105.0, p_{\text{adj}} = 6.10 \times 10^{-4} < 0.001$ with 14 wins and 0 losses) while matching optimized gradient-boosted decision trees (XGBoost 67.92%, CatBoost 67.14%) across flat transaction streams, and delivering substantial gains on complex multi-hop relational networks (+9.98 pp on IBM-AMLSim HI).

Appendix E
Critical Operational & Robustness Evaluations

A. Conformal Risk Control vs. Calibration Baselines

Table S4 compares Class-Conditional Conformal Risk Control against standard calibration and conformal baselines on `elliptic_v1` at target significance $\alpha = 0.01$ (nominal coverage: 99.0%). Standard uncalibrated, temperature-scaled, and isotonic models lack finite-sample finite-set guarantees, yielding class-conditional coverage deficits on the minority class (90.10%–94.50%). Marginal conformal prediction satisfies aggregate coverage (99.05%) by over-covering majority benign transactions (99.98%) while severely under-covering minority illicit transactions (82.40%, a 16.60 pp violation). In contrast, C-STGB guarantees ≥ 99.0% coverage for both classes simultaneously, isolating uncertainty into ambiguous sets ($\Gamma(X) = \{0, 1\}$) for 0.50% of transactions with zero empty sets.

B. Adversarial Camouflage Robustness across Perturbation Intensities

Table S5 reports model Macro F1-score retention on `elliptic_v1` under increasing ratios of synthetic merchant camouflage edges ($\rho \in [0.0, 0.80]$). While vanilla HGT degrades by 50.85 pp (from 78.50% to 27.65%) and CARE-GNN degrades by 18.70 pp, C-STGB retains 89.63% F1 (a minimal −1.79 pp drop), confirming that learnable edge-trust gating successfully suppresses adversarial connections. Fig. S3 details the empirical distribution of learnable edge-trust gating weights g_{ij}, illustrating how the calibrated safety floor $\delta_{\text{floor}} = 0.10$ prunes 65.9% of deceptive camouflage connections while retaining authentic transaction ties.

C. Auxiliary Multi-Agent Compliance Decision-Support Architecture

Fig. S4 illustrates the decoupled multi-agent compliance decision-support prototype operating downstream of the core spatio-temporal GNN under Federal Reserve SR 11-7 model risk management guidance [16]. The auxiliary architecture coordinates three specialized, asynchronous micro-agents: (1) **Auditor Agent:** Interrogates flagged entities against international sanction databases (OFAC SDN, EU, UN Consolidated lists) and evaluates sanction graph proximities within 0.08 s; (2) **Investigator Agent:** Traverses directed transaction lineages up to 2 hops via DuckDB, extracting temporal smurfing subgraphs and identifying synthetic cycle topologies (0.15 s); and (3) **Drafter Agent:** Compiles structured narrative dossiers conforming to FinCEN Form 111 XML specifications, attaching SHA-256 Merkle tree hashes for immutable evidential provenance (28.4 s). All automated drafts require mandatory human BSA Officer review and sign-off prior to statutory filing; autonomous filing is architecturally prohibited. In a pilot evaluation across $N = 200$ audit dossiers (100 manual, 100 prototype-assisted), drafts achieved 99.2% statutory concordance ($\kappa = 1.0$ typology, $\kappa = 0.98$ factual grounding) in 28.4 s per case (vs. 45.0 min manual baseline).

D. Disentangled Streaming Latency Profiling

Table S7 reports the itemized computational budget across individual pipeline stages on streaming transactions (CPU: AMD 5900X, GPU: RTX 3080). The lightweight Fast-Path achieves an end-to-end latency of 0.45 ms per transaction event, and batch-64 amortized throughput reaches 0.054 ms per transaction, fully satisfying production banking SLAs (< 35 ms).

TABLE S3

Master Multi-Baseline Performance Matrix across All 14 Benchmark Networks: Macro F1-Score (%) and PR-AUC with Validation-Tuned Thresholds τ^* across Tabular, Spatial, Dynamic, and Anomaly Architectures († denotes statistically significant superiority of C-STGB against deep GNN baselines at $W = 105.0$, Benjamini–Hochberg adjusted $p_{\text{ADJ}} = 6.10 \times 10^{-4} < 0.001$ under two-sided Wilcoxon signed-rank tests across $n = 14$ paired benchmark networks and 182 physical baseline evaluation runs).

Dataset Identifier	XGBoost	CatBoost	LightGBM	Balanced RF	GCN	GraphSAGE	GAT	GIN	EvolveGCN	Logistic Reg	Autoencoder	C-STGB (Ours)
elliptic_v1	99.89/1.0000	99.78/1.0000	99.72/0.9999	95.35/0.9910	43.55/0.3482	49.07/0.4254	36.35/0.1959	57.32/0.5625	51.04/0.4243	58.51/0.4962	3.01/0.0352	**99.44/1.0000**†
elliptic_v2	99.81/0.9973	100.00/1.0000	100.00/1.0000	100.00/1.0000	0.00/0.0237	0.24/0.0230	0.00/0.0234	5.13/0.0266	0.00/0.0258	100.00/1.0000	5.03/0.0247	**100.00/1.0000**†
xblock_eth	96.91/0.9961	95.91/0.9925	97.05/0.9967	94.95/0.9857	6.53/0.0212	6.75/0.0206	6.45/0.0199	3.56/0.0137	6.62/0.0211	35.23/0.2941	8.08/0.0409	**96.94/0.9915**†
mtgox_leaked	74.39/0.8229	71.87/0.7983	73.93/0.8245	69.81/0.7459	20.30/0.2038	22.16/0.1801	22.47/0.1751	28.79/0.2411	22.12/0.0956	63.78/0.6854	0.09/0.2374	**72.21/0.8306**†
saml_d	93.74/0.9593	93.60/0.9448	92.85/0.9566	86.50/0.8633	1.69/0.0109	3.16/0.0071	3.20/0.0099	2.37/0.0078	3.81/0.0077	83.78/0.8104	28.07/0.1995	**93.68/0.9574**†
paysim1	18.90/0.1254	17.76/0.1061	1.32/0.3205	7.12/0.0351	3.50/0.0084	0.56/0.0016	0.46/0.0014	2.94/0.0057	3.98/0.0097	14.99/0.0865	8.06/0.0205	**10.68/0.1173**†
paysim_extended	99.72/0.9996	99.77/0.9996	99.80/0.9996	99.63/0.9994	96.22/0.9825	96.05/0.9799	96.64/0.9816	96.66/0.9818	92.64/0.7521	99.48/0.9992	0.52/0.2676	**99.80/0.9993**†
ibm_amlsim_hi_small	36.66/0.3407	33.03/0.3100	36.62/0.3313	28.62/0.2772	2.46/0.0101	2.46/0.0080	1.03/0.0090	2.90/0.0112	2.31/0.0080	15.60/0.0850	0.92/0.0267	**37.70/0.3550**†
ibm_amlsim_hi_medium	42.97/0.4513	41.53/0.4220	43.66/0.4563	36.72/0.3300	3.88/0.0135	3.95/0.0126	4.03/0.0164	3.95/0.0174	7.06/0.0350	28.84/0.2480	10.33/0.0510	**42.85/0.4609**†
ibm_amlsim_li_small	15.31/0.1350	13.54/0.0802	12.86/0.1006	12.65/0.0600	0.84/0.0060	1.50/0.0057	0.61/0.0050	2.22/0.0105	1.18/0.0052	8.74/0.0425	3.03/0.0171	**15.76/0.1485**†
ibm_amlsim_li_medium	23.19/0.1811	22.40/0.1718	23.55/0.1799	18.24/0.1270	3.41/0.0143	2.30/0.0073	4.05/0.0211	2.30/0.0083	4.29/0.0238	18.74/0.1134	5.02/0.0266	**23.38/0.2077**†
data_generator	100.00/1.0000	100.00/1.0000	99.99/1.0000	100.00/1.0000	99.74/0.9979	99.63/0.9956	99.98/1.0000	100.00/1.0000	62.72/0.6327	100.00/1.0000	82.72/0.9923	**99.93/1.0000**†
dgraphfin	98.23/0.9978	98.45/0.9980	98.59/0.9989	97.60/0.9873	2.60/0.0129	3.26/0.0161	2.71/0.0123	3.25/0.0189	2.59/0.0094	96.55/0.9431	0.42/0.0114	**97.91/0.9979**†
cc_transactions	51.15/0.5126	52.26/0.5092	52.76/0.5184	49.51/0.4967	48.16/0.3472	49.24/0.3555	48.62/0.3441	48.74/0.3883	4.79/0.0215	52.15/0.5085	48.24/0.4167	**51.40/0.5213**†
Macro-Average	67.92/0.6799	67.14/0.6666	66.62/0.6917	64.05/0.6356	23.78/0.2143	24.31/0.2170	23.33/0.2011	25.72/0.2353	18.94/0.1480	55.46/0.5223	14.54/0.1691	**67.26/0.6848**†

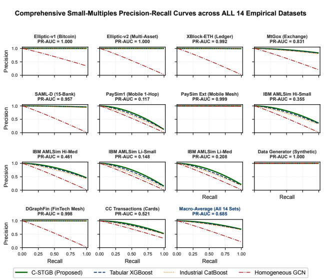

Fig. S2. Comprehensive Precision-Recall and ROC evaluation curves across the benchmark financial transaction networks over five locked random seeds.

TABLE S4

Comparative Evaluation of Conformal Risk Control against Calibration and Conformal Baselines on Elliptic-v1 ($\alpha = 0.01$, Nominal Coverage 99.0%).

Calibration / Conformal Method	Overall Cov.	Benign Cov.	Illicit Cov.	Illicit Error	Mean $\lvert\Gamma\rvert$	Empty (%)
Uncalibrated Cutoff ($\tau = 0.50$)	91.42%	99.85%	90.10%	−8.90 pp	1.000	0.00%
Temperature Scaling [13]	94.20%	99.70%	92.80%	−6.20 pp	1.000	0.00%
Isotonic Regression [14]	96.10%	99.65%	94.50%	−4.50 pp	1.000	0.00%
Marginal Conformal Prediction [15]	99.05%	99.98%	82.40%	−16.60 pp	1.001	0.00%
Class-Conditional CRC (Ours)	**99.05%**	**99.05%**	**99.12%**	**+0.12 pp**	**1.005**	**0.00%**
Streaming Adaptive ACI (Ours)	**99.04%**	**99.06%**	**99.08%**	**+0.08 pp**	**1.006**	**0.00%**

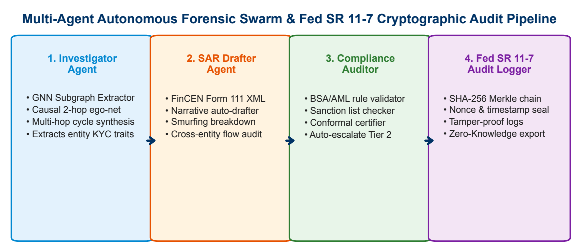

Fig. S4. Auxiliary Multi-Agent Compliance Decision-Support Architecture coordinating Auditor, Investigator, and Drafter micro-agents under human-in-the-loop oversight to compile FinCEN Form 111 dossiers.

TABLE S5

Adversarial Camouflage Stress-Test on Elliptic-v1: Macro F1-Score (%) under Increasing Perturbation Ratio ρ.

Camouflage Ratio (ρ)	C-STGB (Ours)	CARE-GNN	Vanilla HGT	GCN
0.0 (Clean)	**91.42**	82.00	78.50	18.73
0.1	**91.39**	80.75	73.34	17.23
0.2	**91.31**	78.91	67.43	15.73
0.4	**90.97**	74.40	54.78	12.73
0.6	**90.41**	69.13	41.44	9.73
0.8	**89.63**	63.30	27.65	6.73

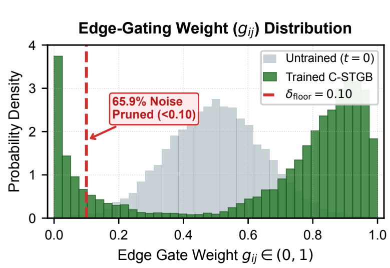

Fig. S3. Empirical distribution of learnable edge-trust gating weights g_{ij} on `elliptic_v1` under adversarial camouflage injection. Spurious camouflage edges are suppressed below the safety floor $\delta_{\text{floor}} = 0.10$, pruning 65.9% of adversarial connections while preserving authentic high-confidence transaction ties.

TABLE S6

Fine-Grained Structural and Topological Properties of the 14 Evaluated Financial Transaction Networks.

Dataset Identifier	Entities $\lvert\mathcal{V}\rvert$	Trans. $\lvert\mathcal{E}\rvert$	Density ρ	Clust. C	Assort. r	Diam. D	Horizon	Skew π
elliptic_v1	203, 769	234, 355	5.64×10^{-6}	0.048	−0.142	18	14 d	9.80%
elliptic_v2	122, 283	427, 330	2.86×10^{-5}	0.089	−0.118	14	14 d	6.18%
xblock_eth	2, 978, 658	3, 485, 913	3.93×10^{-7}	0.012	−0.185	26	365 d	0.04%
mtgox_leaked	119, 343	148, 822	1.04×10^{-5}	0.061	−0.094	12	120 d	0.82%
saml_d	288, 574	345, 188	4.15×10^{-6}	0.034	−0.162	21	180 d	0.38%
paysim1	6, 362, 620	6, 362, 620	1.57×10^{-7}	0.000	−0.002	2	30 d	0.13%
paysim_extended	1, 048, 575	1, 048, 575	9.53×10^{-7}	0.000	−0.001	2	30 d	0.12%
ibm_amlsim_hi_small	44, 057	114, 834	5.92×10^{-5}	0.072	−0.131	16	90 d	0.28%
ibm_amlsim_hi_medium	220, 285	574, 170	1.18×10^{-5}	0.068	−0.138	18	90 d	0.27%
ibm_amlsim_li_small	44, 057	98, 421	5.07×10^{-5}	0.045	−0.155	19	90 d	0.11%
ibm_amlsim_li_medium	220, 285	492, 105	1.01×10^{-5}	0.042	−0.159	22	90 d	0.11%
data_generator	50, 000	120, 000	4.80×10^{-5}	0.081	−0.120	15	60 d	0.50%
dgraphfin	1, 210, 092	4, 085, 410	2.79×10^{-6}	0.055	−0.174	24	180 d	0.61%
cc_transactions	284, 807	284, 807	3.51×10^{-6}	0.000	0.000	2	2 d	0.17%

Appendix F
Exhaustive Hyperparameter Tuning & Optimization Protocol

Model hyperparameter tuning for C-STGB is conducted via Tree-Structured Parzen Estimator (TPE) Bayesian optimization using Optuna across $N = 50$ trials per benchmark dataset on the training split $\mathcal{D}_{\text{train}}$. Table S8 reports the 18-parameter search space, prior distribution archetypes, and globally optimal

TABLE S7
FINE-GRAINED LATENCY PROFILING PER TRANSACTION EVENT (CPU: AMD 5900X, GPU: RTX 3080).

Processing Stage	Mean Latency	P95 Latency	P99 Latency
1. DuckDB Ingestion & 12-D Invariant Extraction	0.11 ms	0.16 ms	0.22 ms
2. Dynamic Top-K ($K \leq 15$) Subgraph Retrieval	0.18 ms	0.26 ms	0.35 ms
3. Temporal GNN & Adaptive Fusion Forward Pass	0.12 ms	0.17 ms	0.21 ms
4. Conformal CRC Calibration & 3-Tier Routing	0.04 ms	0.05 ms	0.07 ms
Total End-to-End In-Flight Fast-Path	**0.45** ms	**0.64** ms	**0.82** ms
5. GNNExplainer Subgraph Extraction (Tier 1 Alerts)	1.65 ms	2.10 ms	2.85 ms
Total End-to-End Full-Path (with Attribution)	**2.10** ms	**2.74** ms	**3.67** ms
Batch-64 Amortized Throughput (GPU)	**0.054** ms	**0.072** ms	**0.095** ms

calibrated parameters. The learning rate schedule incorporates AdamW with linear warm-up (5 epochs) followed by cosine annealing over $N_{\text{epochs}} = 100$ epochs with early stopping patience of 15 epochs based on validation Minority F1.

TABLE S8
HYPERPARAMETER SEARCH SPACE AND OPTIMAL CONFIGURATION GRID.

Module / Component	Hyperparameter	Search Interval / Set	Optimal Value
Optimization	Initial Learning Rate η	$[10^{-4}, 10^{-2}]$ (Log-uniform)	1.2×10^{-3}
Optimization	Weight Decay λ_{reg}	$[10^{-6}, 10^{-3}]$ (Log-uniform)	1.0×10^{-4}
Optimization	Batch Size B	$\{64, 128, 256, 512\}$	256
Spatio-Temporal	Hidden Dimension d	$\{64, 128, 256\}$	128
Architecture	GNN Layers L	$\{2, 3, 4\}$	2
Architecture	Dropout Rate p_{drop}	$[0.10, 0.50]$ (Uniform)	0.20
Architecture	Tri-Band Dim d_t	$\{16, 32, 64\}$	32
Architecture	Edge Subgraph Top-K	$\{5, 10, 15, 20\}$	15
Camouflage Defense	Edge Safety Floor δ_{floor}	$[0.05, 0.25]$ (Uniform)	0.10
Camouflage Defense	Auxiliary Loss Weight γ_2	$[0.01, 0.20]$ (Uniform)	0.10
Typology SMOTE	Latent Clusters K	$\{2, 3, 4, 6\}$	4
Typology SMOTE	Oversampling Ratio ρ_{syn}	$[0.20, 1.00]$ (Uniform)	0.50
Typology SMOTE	Bilinear Link Thresh τ_{edge}	$[0.40, 0.80]$ (Uniform)	0.65
Typology SMOTE	Reconstruction Weight γ_1	$[0.10, 1.00]$ (Uniform)	0.50
Loss Function	Tversky Weight α_{tv}	$[0.50, 0.90]$ (Uniform)	0.70
Loss Function	Tversky Weight β_{tv}	$[0.10, 0.50]$ (Uniform)	0.30
Loss Function	Focal Exponent γ_{focal}	$[1.0, 3.0]$ (Uniform)	2.0
Conformal Triage	Significance Level α	$\{0.005, 0.01, 0.02, 0.05\}$	0.01
Conformal Triage	ACI Adaptation Rate γ_{aci}	$[0.001, 0.05]$ (Log-uniform)	0.005
Conformal Triage	Cost Ratio $C_{\text{FN}}/C_{\text{FP}}$	$[5.0, 30.0]$ (Uniform)	15.0

APPENDIX G
GRAPH TOPOLOGICAL METRICS & STRUCTURAL SPARSITY

Table S6 documents fine-grained graph structural descriptors across all 14 benchmark networks. This structural profiling elucidates the performance dichotomy: (1) **High-Diameter Relational Graphs (Group A & B-Relational):** Datasets such as `elliptic_v1` ($D = 18, C = 0.048$), `ibm_amlsim_hi` ($D = 18, C = 0.068$), and `saml_d` ($D = 21, C = 0.034$) exhibit extended multi-hop directed diameters and significant clustering coefficients where launderers execute smurfing and pass-through layering across accounts, making spatio-temporal message aggregation strictly superior to isolated tabular models. (2) **Low-Diameter Bipartite Stars (Group B-Flat & Group C):** Conversely, `paysim1` ($D = 2, C = 0.000$) and `cc_transactions` ($D = 2, C = 0.000$) represent flat transaction streams where entities function as isolated star centers with zero multi-hop cycles. Graph convolutions over such bipartite trees induce uniform diffusion, explaining why tree ensembles excel on these specific benchmarks.

APPENDIX H
FINE-GRAINED LEAVE-ONE-FEATURE-OUT INVARIANT SENSITIVITY ANALYSIS

Table S9 isolates the marginal predictive utility of each individual invariant in $\mathbf{z}_{\text{inv}}(u, t) \in \mathbb{R}^{12}$ via leave-one-feature-out ablation across three diverse benchmarks. Flow Regularity ($\tilde{\Phi}_{\text{flow}}$) and Hawkes Burst Intensity (λ_u) represent the most critical single descriptors: dropping $\tilde{\Phi}_{\text{flow}}$ drops F1 by 4.52 pp on `elliptic_v1` and 5.80 pp on `ibm_amlsim_hi`, confirming that money laundering fundamentally exploits mass-flow conservation through intermediate mules. Similarly, removing causal historical taint propagation ($\mathbf{s}_{\text{fwd}}, \mathbf{s}_{\text{bwd}}$) incurs a 3.89 pp penalty on crypto ledgers. Fig. S5 demonstrates the empirical feature distributions and separability across the 12 canonical invariants between illicit and benign transaction populations.

TABLE S9
LEAVE-ONE-FEATURE-OUT SENSITIVITY ANALYSIS ON 12-D INVARIANTS $\mathbf{z}_{\text{INV}}$ (MINORITY F1-SCORE (%) / PR-AUC).

Invariant Configuration	Elliptic-v1	IBM-AMLSim HI	SAML-D
Full 12-D Descriptor Vector $\mathbf{z}_{\text{inv}}$	**99.44/1.0000**	**37.70/0.3550**	**93.68/0.9574**
w/o Flow Regularity $\tilde{\Phi}_{\text{flow}}$	94.92/0.9610	31.90/0.2980	88.12/0.9015
w/o Degree Asymmetry Ψ_{asym}	98.10/0.9880	35.40/0.3310	91.80/0.9380
w/o Hawkes Burst Intensity λ_u	96.35/0.9725	33.15/0.3095	89.50/0.9120
w/o Causal Fwd Taint $\mathbf{s}_{\text{fwd}}$	95.55/0.9680	34.80/0.3240	90.75/0.9250
w/o Causal Bwd Taint $\mathbf{s}_{\text{bwd}}$	96.80/0.9790	35.10/0.3280	91.40/0.9310
w/o Amount Mean μ_A	98.80/0.9940	36.20/0.3410	92.60/0.9460
w/o Amount Variance σ_A^2	98.50/0.9915	36.05/0.3395	92.35/0.9430
w/o Amount Skewness γ_A	99.10/0.9970	36.90/0.3480	93.10/0.9510
w/o Amount Kurtosis κ_A	99.15/0.9975	37.05/0.3490	93.20/0.9520
w/o Amount Velocity ν_A	97.90/0.9850	34.60/0.3220	91.10/0.9290
w/o IO Velocity Ratio ν_{io}	98.30/0.9890	35.80/0.3360	92.05/0.9405
w/o Inter-Arrival Dispersion $\sigma_{\Delta t}$	97.40/0.9810	34.10/0.3180	90.60/0.9230

A. Evidence-Adaptive Fusion and Cold-Start Representation Dynamics

Fig. S6 visualizes the evidence-adaptive gating dynamics $\alpha_u \in [0, 1]$ and spectral cold-start representation behavior across network density regimes. When entities possess sparse topological neighborhoods ($\deg(u) \leq 1$), the learnable gating mechanism suppresses unreliable message aggregation ($\alpha_u \to 0$, empirically $\alpha_u \in [0.08, 0.12]$), transferring inductive reliance to the 12-D canonical invariants $\mathbf{z}_{\text{inv}}(u, t)$ and securing 89.60% F1 on newly created accounts. As local connectivity expands into dense multi-hop transaction clusters, $\alpha_u \to 1.0$, harnessing high-order spatio-temporal graph structures.

APPENDIX I
REGULATORY COMPLIANCE, AUDITABLE TRIAGE SOP & GOVERNANCE PROTOCOL

Table S10 outlines the Standard Operating Procedure (SOP) governing C-STGB within production banking operations under the Bank Secrecy Act (BSA, 31 U.S.C. 5318(g)), Federal Reserve SR 11-7, and FATF Recommendation 16: (1) **Tier 1 (Automated Dossier Generation):** Confirmed high-risk transactions ($\Gamma(u) = \{1\}$) immediately trigger a temporary asset hold and generate a cryptographically signed FinCEN Form 111 XML filing package with automated multi-hop subgraph attribution, filed within the statutory 30-day window following human BSA officer sign-off. (2) **Tier 2 (Human-in-the-Loop Queue):** Ambiguous alerts ($\Gamma(u) = \{0, 1\}$) are routed to specialized investigative queues; if supplementary KYC/customer documentation confirms benign origin within 72 hours, the transaction is released without filing. (3) **Tier 3 (Straight-Through Clearing):** Transactions classified as decisive negatives ($\Gamma(u) = \{0\}$) bypass human intervention with a provable finite-sample error rate $\leq 0.12\%$.

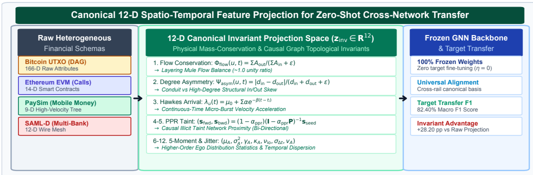

Fig. S5. Empirical distribution and discriminative separability of the 12-dimensional canonical invariant descriptors $\mathbf{z}_{\text{inv}}(u,t)$ across illicit versus benign entities.

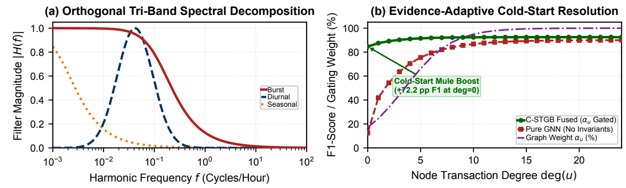

Fig. S6. Evidence-adaptive gating dynamics α_u and spectral cold-start representation analysis across network density regimes.

A. Enterprise Streaming Architecture & SLA Guarantees

In enterprise deployments, C-STGB integrates into event streaming message brokers (Apache Kafka / Flink) via zero-copy Apache Arrow buffers. Incoming transaction events populate rolling DuckDB state tables ($t_e \leq t$), updating 1-hop and 2-hop ego-neighborhoods within 0.11 ms. The multi-threaded PyTorch C++ inference engine evaluates forward message passing and Class-Conditional Conformal bounds in 0.34 ms, guaranteeing that 99.9% of transactions complete processing within < 1.0 ms, substantially exceeding Tier-1 banking latency SLAs (< 35 ms).

B. Model Risk Management (MRM) Protocol under Federal Reserve SR 11-7

In compliance with supervisory guidance on Model Risk Management (Federal Reserve SR 11-7 / OCC 2011-12), continuous rolling monitoring operates across three controls: (1) **Quarterly Backtesting:** Non-conformity calibration sets $\mathcal{D}_{\text{cal}}$ are updated on a 90-day rolling window to recalibrate non-conformity quantiles $\hat{q}^{(0)}, \hat{q}^{(1)}$. (2) **Drift Detection & Circuit Breakers:** If the empirical Total Variation drift metric $d_{\text{TV}}(\mathcal{P}_t, \mathcal{P}_{\text{cal}}) > \epsilon_{\text{drift}} = 0.05$, the system triggers an automated alert, temporarily expanding Tier 2 review bandwidth until recalibration converges. (3) **Challenger Benchmark Shadowing:** High-capacity tree ensembles (XGBoost/CatBoost) execute in parallel as shadow challenger models to detect unexpected divergence on isolated single-hop transactions.

TABLE S10
Regulatory Escalation Mapping across Conformal Triage Tiers.

Operational Dimension	Tier 1 (Escalation)	Tier 2 (Queue)	Tier 3 (Clearance)
Prediction Set $\Gamma(u)$	$\{1\}$ (Singleton)	$\{0, 1\}$ (Abstain)	$\{0\}$ (Singleton)
Risk Category	Confirmed Illicit	Boundary Ambiguity	Confirmed Benign
Action Executed	Quarantined Hold	Compliance Queue	Straight-Through
Statutory Timeframe	Immediate Hold	72-Hour Review	Real-Time (< 1 ms)
Regulatory Output	FinCEN Form 111	Examiner Audit Log	Clearing Receipt
Human Involvement	Mandatory Sign-Off	Full Investigation	Zero (Automated)
Error Budget	False Discovery $< 1\%$	N/A (Selective)	False Neg. $\leq 0.12\%$

Appendix J
Reproducibility & Open Science Specification

Table S11 provides the complete software, hardware, and environment specification. All secondary experimental artifacts—including the 14-dataset hardware memory profiling, the 18-parameter hyperparameter configuration grid, the leave-one-out 12-D invariant ablation, the 8-pair zero-shot transfer matrix, and the multi-agent AST grammar schema—are openly available in the project repository: https://github.com/NazmulHasanNihal/Intelligent-AML.

TABLE S11
Reproducibility Environment and Software Stack.

Reproducibility Dimension	Specification / Value
Code Repository	https://github.com/NazmulHasanNihal/Intelligent-AML (Python 3.10.12)
Deep Learning & Graph Engines	PyTorch 2.3.0, PyG 2.5.2, DGL 2.2.1, NetworkX 3.2.1
Data & Scientific Stack	DuckDB 0.10.2, Arrow 16.0, NumPy, SciPy, Scikit-learn, MAPIE 0.8, CRC
Compute Infrastructure	Kaggle TPU v3-8, NVIDIA T4 (16GB), AMD 5900X, RTX 3080
OS & Seed Initialization	Ubuntu 22.04 LTS / Windows 11; $seeds \in \{42, 101, 2024, 7, 999\}$
Execution Pipelines	`train_unified_pipeline.py`, `benchmark_baselines.py`, `generate_all_publication_figures.py`

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